

Building intelligent Global Health and
Humanitarian Response Technologies

BLOCK II:

High Stakes Decisions

LiGHT

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EPFL

ARIADNE LABS


ASHOKA
UNIVERSITY

**WORLD
ECONOMIC
FORUM**

Rwanda
Centre for the
Fourth Industrial
Revolution

When an algorithm is right, why might a clinician still be right not to use it?

- A model can outperform experts on cases and still harm patients in practice.
- Clinical performance is not clinical usefulness.
- Every recommendation enters a system of incomplete records, interruptions, liability, hierarchy, trust, and exhausted people.
- In medicine, the final benchmark is not “Was the answer correct?” It is: Did care become safer, fairer, and better?

The Machine That Was Right and *Was Never Used*

Lecturer: *Prof. Annie Hartley*

Light

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Outline

01 1979: MYCIN

02 Accuracy vs Reality

03 CDSS

04 XXX

05 XXX

WEEK 5: Theme: Why has the best medical AI never treated a patient?

MYCIN outperformed every Stanford faculty physician in 1979. No patient was ever treated by it. This week we study the fifty-year gap between clinical AI performance and clinical AI deployment — and why understanding that gap is the prerequisite for closing it.

CONTEXT

SECTION	FIELD	CONTENT
OVERVIEW	Block	Block II – High-Stakes Decisions (Weeks 5–8)
	Week	Week 5 – Why Has the Best Medical AI Never Treated a Patient?
	Central Question	Why has the best medical AI never treated a patient?
	Red Thread	Trust
	Learning Objective	Understand why high-performing AI often fails in clinical deployment.
DOMAIN LECTURE (45 min)	Title	The Machine That Was Right and Was Never Used
	Big Idea	Expert Systems
	Core Concepts	- Expert systems - Knowledge acquisition bottleneck - Five rights of CDS - Certainty factors - MYCIN - INTERNIST-1 - HELP - UpToDate - Liability barrier - Knowledge currency
	Great Debate	Should benchmark performance alone justify deployment?
	Historical Story	Why MYCIN never reached patients.
	Great Thinker	Edward Shortliffe
	Memorable Fact	MYCIN outperformed experts but treated no patients.
	Read Before Dying	Topol, E.J. (2012). The Creative Destruction of Medicine. Basic Books.

	Seminal Papers	Ledley & Lusted (1959). Reasoning Foundations of Medical Diagnosis. Science. Yu et al. (1979). Antimicrobial Selection by a Computer. JAMA. Bates et al. (2003). Ten Commandments for Effective CDS. JAMIA.
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Mycin

In a famous 1979 evaluation, infectious-disease experts judged MYCIN's therapy recommendations acceptable in 90.9% of cases. It performed at least comparably with expert physicians in the paper test.

And yet:

MYCIN never treated a patient.

Mycin why not?

Eval Gap:

MYCIN was judged on carefully assembled case summaries.

But real clinical work contains:

- missing and contradictory data;
- uncertain diagnoses;
- delayed laboratory results;
- changing patient status;
- competing priorities;
- interruptions;
- patients who cannot follow the recommended plan.

A model can be correct *conditional on clean inputs* and still be unsafe in the world that supplies dirty ones.

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workflow Gap:

A recommendation is useful only if it arrives:

- to the **right person**;
- with the **right information**;
- in the **right format**;
- through the **right channel**;
- at the **right time**.

The liability gap

If MYCIN gives harmful advice:

- Is the clinician responsible for following it?
- Is the clinician responsible for ignoring it?
- Is the hospital responsible for installing it?
- Is the developer responsible for its logic?
- What happens when the algorithm was trained elsewhere but deployed here?

Clinical AI does not eliminate responsibility. It redistributes responsibility—often without admitting it.

4. The local-validity gap

Mycin why not?

INTERNIST-1

A broad diagnostic expert system intended to reason across internal medicine.

- Ambitious, encyclopaedic, and computationally demanding.
- Important legacy: showed that breadth creates a combinatorial problem.
- Lesson: “more medical knowledge” does not automatically produce a usable clinical tool.

HELP

A hospital-based decision-support system developed at LDS Hospital.

- Generated alerts, reminders, and decision support inside care delivery.
- Lesson: integration matters. The system has to meet clinicians where work already happens.

UpToDate

Not usually called “AI,” but perhaps the most successful form of clinical decision support.

- It does not pretend to replace clinical judgement.
- It succeeds because clinicians can pull information when they need it.
- Lesson: a tool that supports professional agency may be adopted more readily than one that appears to overrule it.

The uncomfortable contrast

MYCIN may have been more technically dazzling.
UpToDate may have been more clinically deployable.

Automation bias

The tendency to accept a computer recommendation too readily, even when it is wrong. → cite breast cancer study with radiologists + cognitive atrophy

Algorithm aversion

The tendency to reject an algorithm after observing even a small error—often more harshly than a human making the same error.

Alert fatigue

When too many warnings train clinicians to ignore warnings, including important ones.

Calibration

Whether a predicted probability means what it says. A model that says “80% risk” should be correct roughly 80% of the time in the population where it is used.

Dataset shift

When the world changes from the setting in which the model was trained: a new hospital, patient group, clinical practice, pathogen, or documentation style.

Algorithmic friction

The extra work an AI tool creates: logging in, correcting data, explaining decisions, responding to alerts, documenting overrides.

Silent failure

A model continues producing confident outputs after its assumptions have become false.

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Human vigilance declines because the system is assumed to be watching.

Key distinction

Questions?

From Technological *Implementation* to Clinical *Use*

Lecturer: *Dr. David Sasu*

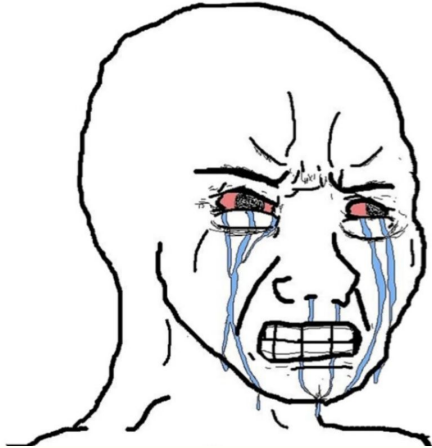
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Outline

- 01 IBM Watson for Oncology
- 02 Why a CDSS fails
- 03 How we design a CDSS that earns human trust
- 04 Engineering robustness into your tools
- 05 Back to your Projects

Looks cool right?

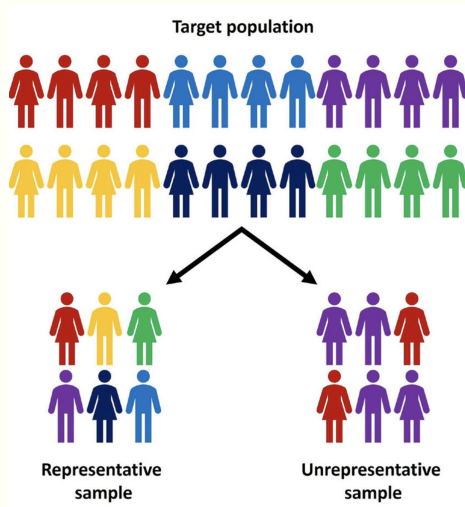


The screenshot shows a website header with a mountain logo, navigation links for 'Services', 'Research', and 'About me', a search icon, and social media links for LinkedIn and a flag icon, along with a 'Contact' button. The main title of the article is 'Case Study 20: The \$4 Billion AI Failure of IBM Watson for Oncology'. Below the title, it says 'Alle Posts' and '7. December 2024'. The article text begins with 'The scale of the ambition became visible in 2015, when IBM formally launched Watson Health as a dedicated business unit and began assembling capabilities through a series of acquisitions that transformed the initiative from a product into a strategic platform. The company acquired Explorys and Phytel, paid \$1 billion for Merge Healthcare, and announced the \$2.6 billion acquisition of...'. On the right side, there is a 'Tags' section with a red-bordered box containing the text 'Technology Transformation'.

<https://www.henricodolfing.ch/en/case-study-20-the-4-billion-ai-failure-of-ibm-watson-for-oncology/>

Why IBM Watson for Oncology failed . . .

1. Unrepresentative training data and poor generalization across healthcare settings



Why IBM Watson for Oncology failed . . .

1. Unrepresentative training data and poor generalization across healthcare settings
2. Unreliable clinical/health recommendations



WARNING:
VIEWER DISCRETION
IS ADVISED.

<https://nypost.com/2025/10/24/health/real-life-ways-bad-advice-from-ai-is-sending-people-to-the-er/>

Why IBM Watson for Oncology failed . . .

1. Unrepresentative training data and poor generalization across healthcare settings
2. Unreliable clinical/health recommendations
3. Scaling before sufficient real-world validation.



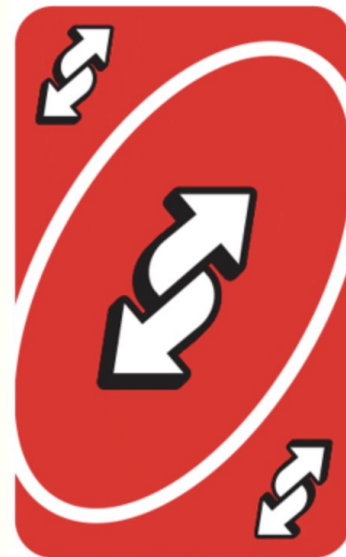
Why IBM Watson for Oncology failed . . .



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Why IBM Watson for Oncology failed . . .

1. Unrepresentative training data and poor generalization across healthcare settings
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How do we design a CDSS that earns a clinician's trust?

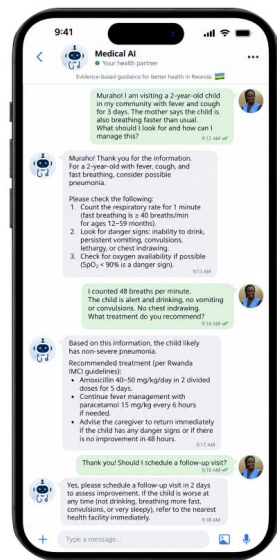


1. **Representative** training data and generalization across healthcare settings
2. **Reliable** clinical recommendations
3. **Real-world validation** before scaling.

Engineering approaches to increasing clinician trust in AI CDSS

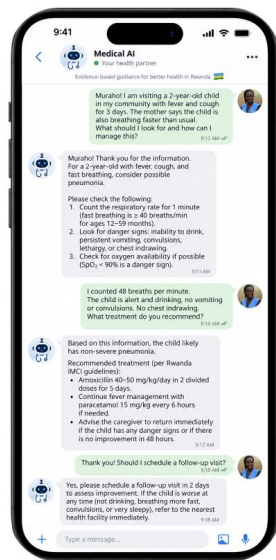
Let's assume that the goal is ...

To build an AI-powered medical assistant for health workers in Rwanda ...



Let's assume that the goal is ...

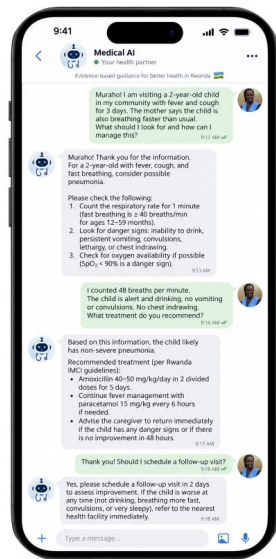
To build an AI-powered medical assistant for health workers in Rwanda ...



Problem 1: Is the underlying model well trained? And if not what do you do?

Let's assume that the goal is ...

To build an AI-powered medical assistant for health workers in Rwanda ...

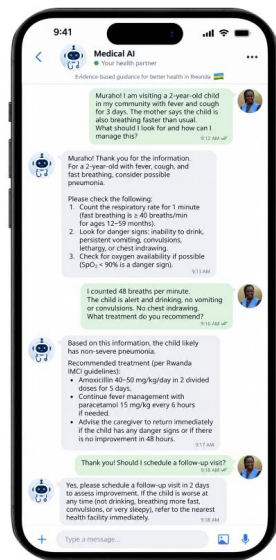


Problem 1: Is the underlying model well trained? And if not what do you do?

Problem 2: How do you ensure that the answers are not hallucinated?

Let's assume that the goal is ...

To build an AI-powered medical assistant for health workers in Rwanda ...

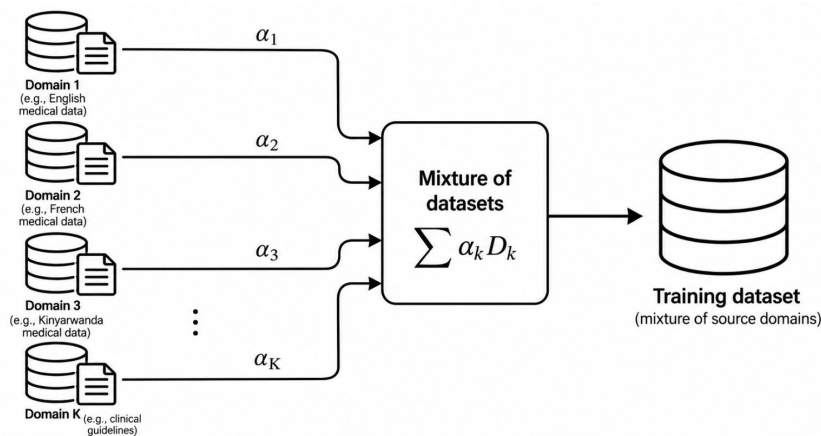


Problem 1: Is the underlying model well trained? And if not what do you do?

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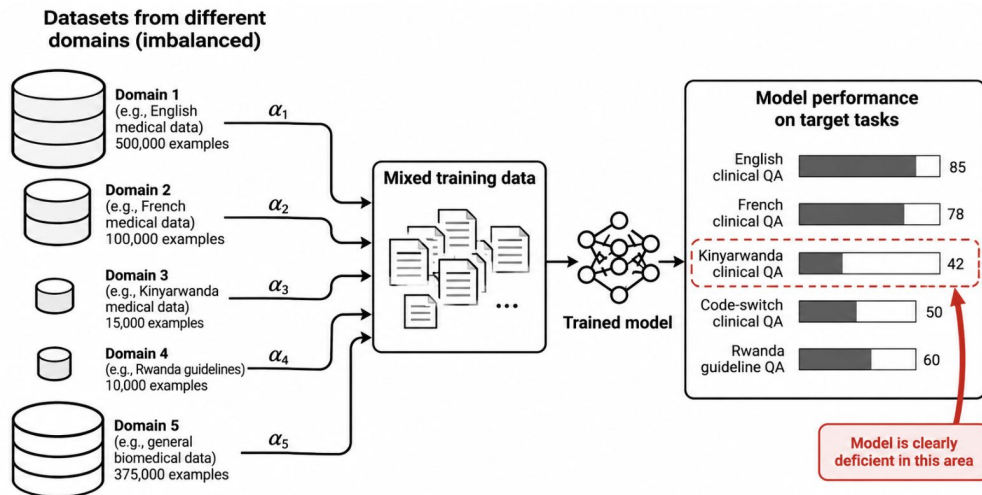
Problem 3: How do you verify the answers that are given to ensure they are true and safe?

Training time problem



If a model sees much more data from one domain than another, the gradients produced by that domain will have more influence on how the model parameters change.

Training time problem



If a model sees much more data from one domain than another, the gradients produced by that domain will have more influence on how the model parameters change.

The choice of data mixture can therefore change which downstream abilities the model learns well.

Training time problem

Dataset domain	Example contents	Dataset size
D1	English medical QA	500,000
D2	French medical QA	100,000
D3	Kinyarwanda medical QA	15,000
D4	Rwanda clinical guidelines	10,000
D5	General biomedical guidelines and instructions	375,000

Question: If we were to simply concatenate this and train normally, what do you think will determine what the model learns?

Training time problem

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D4	Rwanda clinical guidelines	10,000
D5	General biomedical guidelines and instructions	375,000

Question: If we were to simply concatenate this and train normally, what do you think will determine what the model learns?

Answer: Our model will receive far more parameter updates from English / general medical domains than by the small Kinyarwanda / Rwanda specific domains

Training time problem

What we can do about this ...

GRAPE 🍇: Optimize Data Mixture for Group Robust Multi-target Adaptive Pretraining

Simin Fan, Maria Ios Glarou, Martin Jaggi
EPFL
firstname.lastname@epfl.ch

Abstract

The performance of large language models (LLMs) across diverse downstream applications is fundamentally governed by the quality and composition of their pretraining corpora. Existing **domain reweighting algorithms** primarily optimize data mixtures for a single target task, thereby resulting in models that overfit to specialized objectives while exhibiting substantial performance degradation on other benchmarks. This paper introduces **Group Robust Multi-target Adaptive Pretraining (GRAPE)**, a novel multi-source-multi-target domain reweighting framework designed to calibrate pretraining data mixtures for robust performance across multiple target tasks simultaneously. GRAPE dynamically adjusts sampling weights across source domains (*domain weights*) while concurrently modulating *task weights* that quantify the relative importance of each individual target task. This adaptive process prioritizes tasks based on their learning difficulty throughout training. **We formulate this interleaved reweighting mechanism as a minimax optimization problem:** The inner maximization adjusts task weights leveraging **group distributed-robust-optimization (DRO)**, where those tasks demonstrating the least improvement under the current data mixture are prioritized with **higher weights**; The outer minimization then optimizes domain weights to **maximize loss reduction on the prioritized tasks**. Experiments on **ClimbLab** and **SlmPajama** datasets demonstrate that GRAPE consistently outperforms baseline methods in terms of reasoning performance across 6 benchmarks. Furthermore, when applied to multilingual targets, GRAPE effectively identifies optimal training mixtures from mainstream languages, achieving superior language modeling capabilities

Core idea: Once you identify your source data domains and target tasks, during your training process, **assess which target capabilities** are learning more slowly than others and then **increase training exposure to the source domains** whose gradients are useful for improving the lagging capabilities.

5.20380v1 [cs.LG] 26 May 2025

Training time problem

Applying this idea to our initial goal ...

Source data domains	Target tasks	Sampling proportions
English medical QA	English clinical QA	0.40
French medical QA	French clinical QA	0.10
Kinyarwanda medical QA	Kinyarwanda clinical QA	0.05
Rwanda clinical guidelines	Code-switched QA	0.05
General biomedical guidelines and instructions	Rwanda guidelines QA	0.40

Training time problem

Applying this idea to our initial goal ...

Source data domains	Target tasks	Sampling proportions
English medical QA	English clinical QA	0.40
French medical QA	French clinical QA	0.10
Kinyarwanda medical QA	Kinyarwanda clinical QA	0.05
Rwanda clinical guidelines	Code-switched QA	0.05
General biomedical guidelines and instructions	Rwanda guidelines QA	0.40

training →

Loss 1
1.00
1.20
2.00
1.80
1.60

AVG 1.52

Training time problem

Applying this idea to our initial goal ...

Source data domains	Target tasks	Sampling proportions
English medical QA	English clinical QA	0.40
French medical QA	French clinical QA	0.10
Kinyarwanda medical QA	Kinyarwanda clinical QA	0.05
Rwanda clinical guidelines	Code-switched QA	0.05
General biomedical guidelines and instructions	Rwanda guidelines QA	0.40

training →

Loss 1	Loss 2
1.00	0.80
1.20	1.02
2.00	1.96
1.80	1.76
1.60	1.52

AVG 1.52

1.412

Training time problem

Applying this idea to our initial goal ...

Source data domains	Target tasks	Sampling proportions
English medical QA	English clinical QA	0.40
French medical QA	French clinical QA	0.10
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training →

Loss 1	Loss 2
1.00	0.80
1.20	1.02
2.00	1.96
1.80	1.76
1.60	1.52

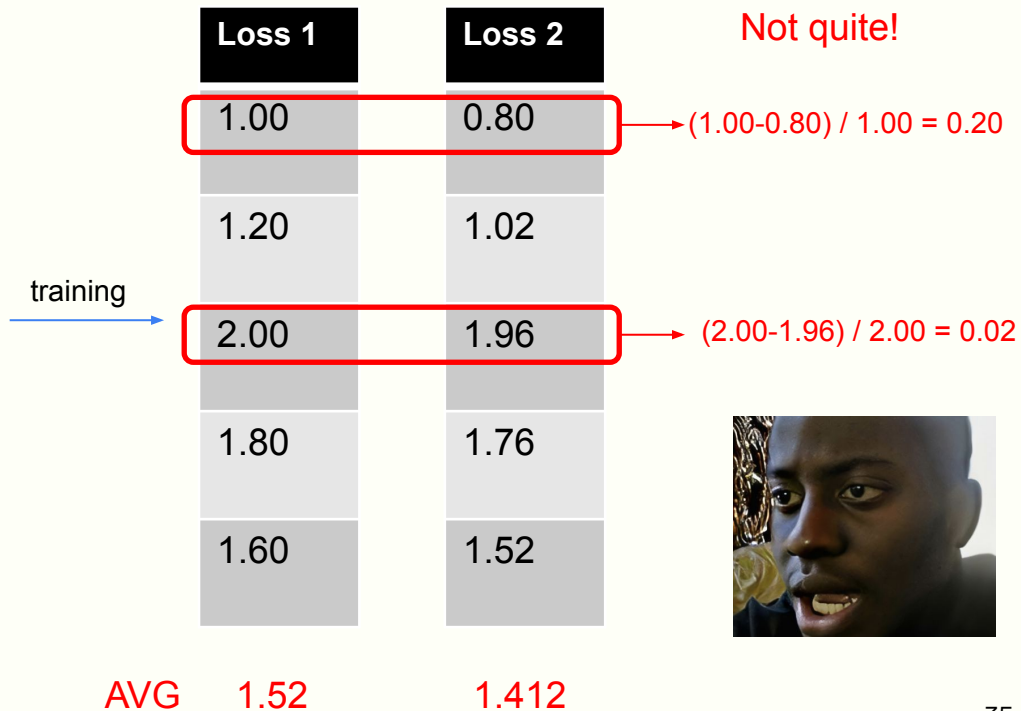
Overall average loss seems to be reducing so everything is cool right?



Training time problem

Applying this idea to our initial goal ...

Source data domains	Target tasks	Sampling proportions
English medical QA	English clinical QA	0.40
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Rwanda clinical guidelines	Code-switched QA	0.05
General biomedical guidelines and instructions	Rwanda guidelines QA	0.40



Training time problem

Applying this idea to our initial goal ...

Target tasks	Formulated task weights
English clinical QA	0.05
French clinical QA	0.10
Kinyarwanda clinical QA	0.40
Code-switched QA	0.30
Rwanda guidelines QA	0.15

Core idea: Once we have identified the tasks that are improving more slowly compared to others, we place greater emphasis on those tasks by applying higher weights to them.

Training time problem

Applying this idea to our initial goal ...

Source data domains	Sampling proportions
English medical QA	0.40
French medical QA	0.10
Kinyarwanda medical QA	0.05
Rwanda clinical guidelines	0.05
General biomedical guidelines and instructions	0.40

Question: Which training domain produces parameter changes that would actually help the lagging target tasks?

Training time problem

Applying this idea to our initial goal ...

Source data domains	Alignment with Kinyarwanda QA
English medical QA	0.08
French medical QA	0.03
Kinyarwanda medical QA	0.71
Rwanda clinical guidelines	0.52
General biomedical guidelines and instructions	0.14

Question: Which training domain produces parameter changes that would actually help the lagging target tasks?

To answer this question, we compare the alignment between the data domain gradients and target task gradients

Training time problem

Applying this idea to our initial goal ...

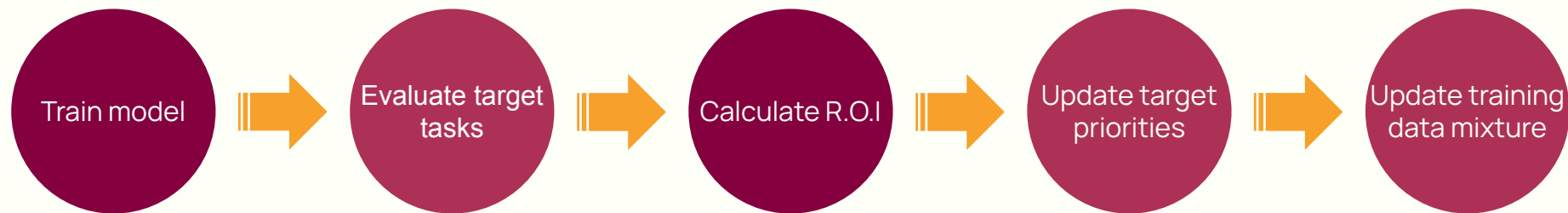
Source data domains	Target tasks	New Sampling Proportions
English medical QA	English clinical QA	0.25
French medical QA	French clinical QA	0.08
Kinyarwanda medical QA	Kinyarwanda clinical QA	0.25
Rwanda clinical guidelines	Code-switched QA	0.20
General biomedical guidelines and instructions	Rwanda guidelines QA	0.22

Old Sampling Proportions
0.40
0.10
0.05
0.05
0.40

After observing the Kinyarwanda and code-switched QA are lacking and observing that Kinyarwanda / Rwanda-specific training gradients helps those targets, we have to readjust our sampling weights to reflect this.

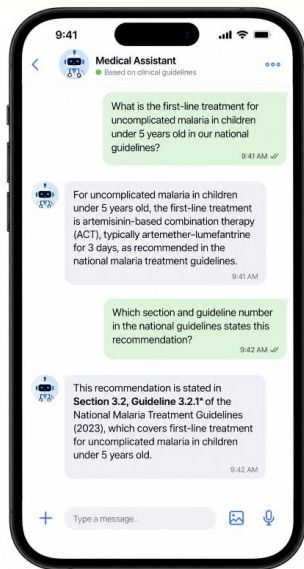
Training time problem

The entire cycle



Inference-time knowledge problem

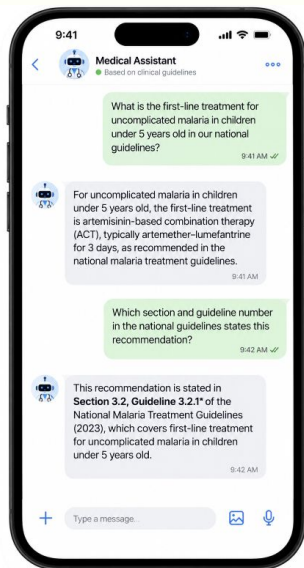
1.



Hallucination

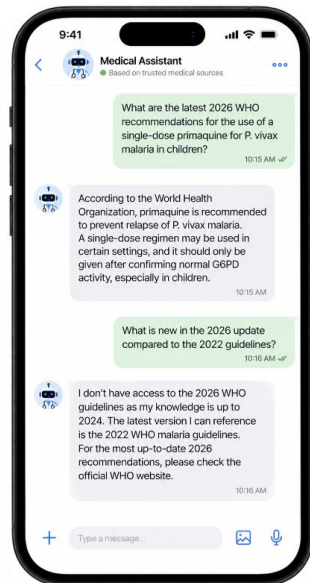
Inference-time knowledge problem

1.



Hallucination

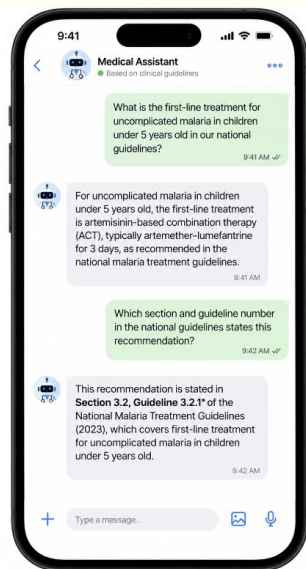
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Knowledge cut-offs

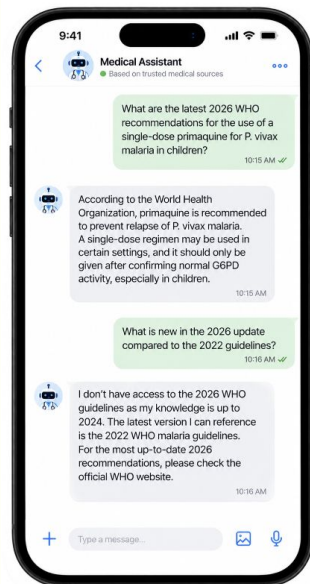
Inference-time knowledge problem

1.



Hallucination

2.

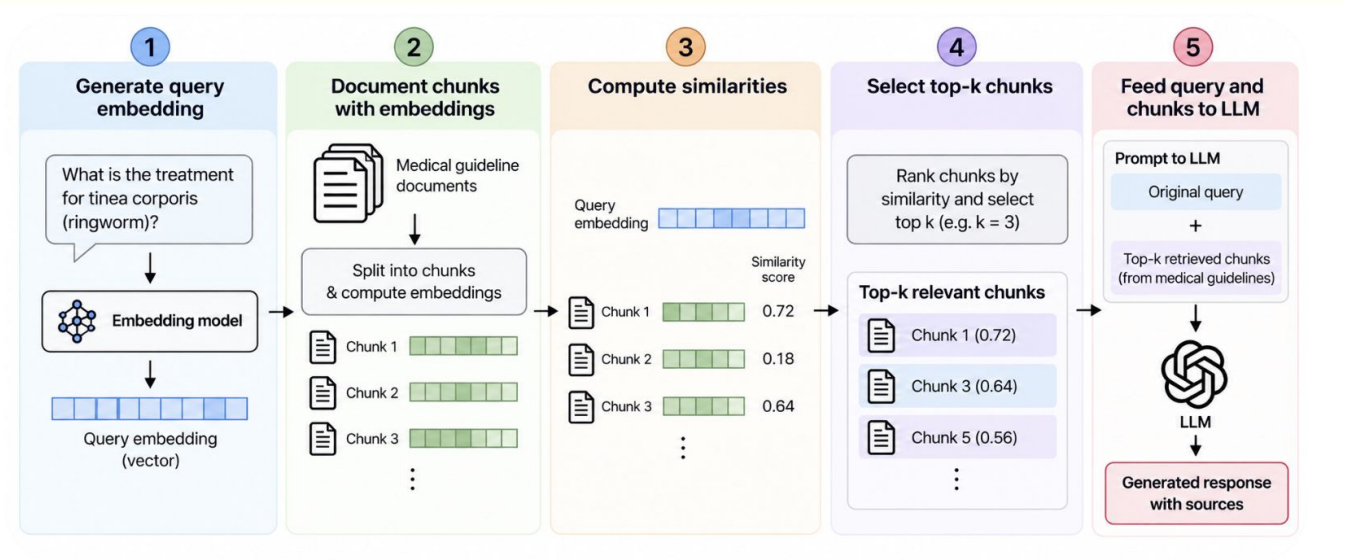


Knowledge cut-offs

Retrieval Augmented Generation (RAG)

Inference-time knowledge problem

How Retrieval Augmented Generation (RAG) works



Inference-time knowledge problem

How do we know RAG works well especially for medical use-cases?

Benchmarking Retrieval-Augmented Generation for Medicine

Guangzhi Xiong^{*†}, Qiao Jin^{♡†}, Zhiyong Lu^{♡§}, Aidong Zhang^{*§}

^{*} University of Virginia

[♡] National Library of Medicine, National Institutes of Health

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Abstract

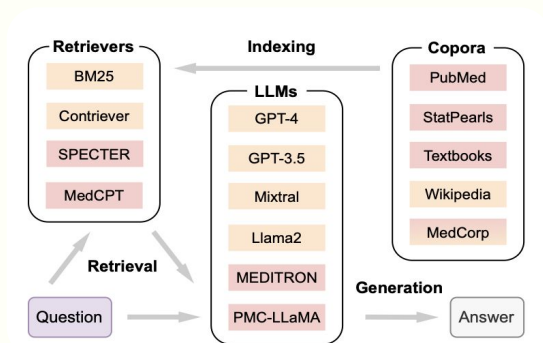
While large language models (LLMs) have achieved state-of-the-art performance on a wide range of medical question answering (QA) tasks, they still face challenges with hallucinations and outdated knowledge. Retrieval-augmented generation (RAG) is a promising solution and has been widely adopted. However, a RAG system can involve multiple flexible components, and there is a lack of best practices regarding the optimal RAG setting for various medical purposes. To systematically evaluate

2023; Anil et al., 2023; Touvron et al., 2023b; Singhal et al., 2023a; Nori et al., 2023a), LLMs often generate plausible-sounding but factually incorrect responses, commonly known as hallucination (Ji et al., 2023). Also, the training corpora of LLMs might not include the latest knowledge, such as recent updates of clinical guidelines. These issues can be especially dangerous in high-stakes domains such as healthcare (Tian et al., 2024; Hersh, 2024).

By providing LLMs with relevant documents retrieved from up-to-date and trustworthy collections,

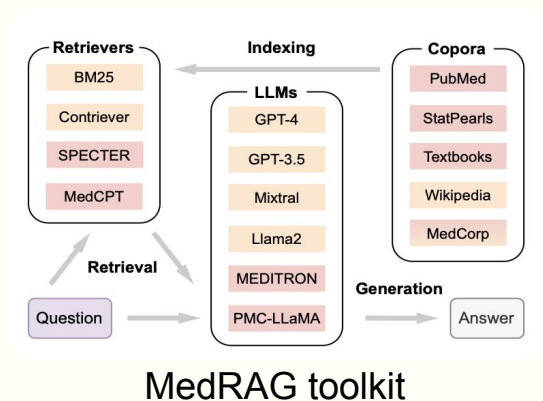
23 Feb 2024

Inference-time knowledge problem



MedRAG toolkit

Inference-time knowledge problem

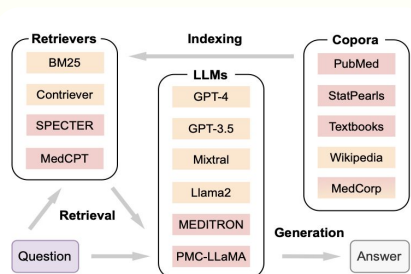


evaluated on

5 Datasets	7,663 Questions	2-4 Choices
MMLU-Med	Which of the following best describes ... ?	A / B / C / D
MedQA-US	A 72-year-old man comes to the physicians ... ?	A / B / C / D
MedMCQA	Axonal transport is:	A / B / C / D
PubMedQA*	Is anorectal endosonography valuable ... ?	Yes / No / Maybe
BioASQ-Y/N	Is medical hydrology the same as Spa ... ?	Yes / No

Medical Information Retrieval-Augmented Generation Evaluation (MIRAGE)

Inference-time knowledge problem



MedRAG toolkit

evaluated on

5 Datasets	7,663 Questions	2-4 Choices
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BioASQ-Y/N	Is medical hydrology the same as Spa ... ?	Yes / No

produced

Medical Information Retrieval-Augmented Generation Evaluation (MIRAGE)

LLM	Method	MIRAGE Benchmark Dataset					Avg.
		MMLU-Med	MedQA-US	MedMCQA	PubMedQA*	BioASQ-Y/N	
GPT-4 (-32k-0613)	CoT	89.44 ± 0.93	83.97 ± 1.03	69.88 ± 0.71	39.60 ± 2.19	84.30 ± 1.46	73.44
	MEDRAG	87.24 ± 1.01	82.80 ± 1.06	66.65 ± 0.73	70.60 ± 2.04	92.56 ± 1.06	79.97
GPT-3.5 (-16k-0613)	CoT	72.91 ± 1.35	65.04 ± 1.34	55.25 ± 0.77	36.00 ± 2.15	74.27 ± 1.76	60.69
	MEDRAG	75.48 ± 1.30	66.61 ± 1.32	58.04 ± 0.76	67.40 ± 2.10	90.29 ± 1.19	71.57
Mixtral (8x7B)	CoT	74.01 ± 1.33	64.10 ± 1.34	56.28 ± 0.77	35.20 ± 2.14	77.51 ± 1.68	61.42
	MEDRAG	75.85 ± 1.30	60.02 ± 1.37	56.42 ± 0.77	67.60 ± 2.09	87.54 ± 1.33	69.48
Llama2 (70B)	CoT	57.39 ± 1.50	47.84 ± 1.40	42.60 ± 0.76	42.20 ± 2.21	61.17 ± 1.96	50.24
	MEDRAG	54.55 ± 1.51	44.93 ± 1.39	43.08 ± 0.77	50.40 ± 2.24	73.95 ± 1.77	53.38
MEDITRON (70B)	CoT	64.92 ± 1.45	51.69 ± 1.40	46.74 ± 0.77	53.40 ± 2.23	68.45 ± 1.87	57.04
	MEDRAG	65.38 ± 1.44	49.57 ± 1.40	52.67 ± 0.77	56.40 ± 2.22	76.86 ± 1.70	60.18
PMC-LLaMA (13B)	CoT	52.16 ± 1.51	44.38 ± 1.39	46.55 ± 0.77	55.80 ± 2.22	63.11 ± 1.94	52.40
	MEDRAG	52.53 ± 1.51	42.58 ± 1.39	48.29 ± 0.77	56.00 ± 2.22	65.21 ± 1.92	52.92

RAG can increase the performance of models

Inference-time knowledge problem

However, BEWARE . . .

The **corpus** as well as the **retrieval approach** can affect performance

Corpus	Retriever	MIRAGE Benchmark Dataset					Average
		MMLU-Med	MedQA-US	MedMCQA	PubMedQA*	BioASQ-Y/N	
None	None	72.91 ± 1.35	65.04 ± 1.34	55.25 ± 0.77	36.00 ± 2.15	74.27 ± 1.76	60.69
PubMed (23.9M)	BM25	72.27 ± 1.36	63.71 ± 1.35	55.49 ± 0.77	66.20 ± 1.15	88.51 ± 1.38	69.23
	Contriever	71.72 ± 1.35	63.94 ± 1.35	54.29 ± 0.77	65.60 ± 1.15	85.44 ± 1.42	68.20
	SPECTER	73.19 ± 1.34	65.20 ± 1.34	53.12 ± 0.77	54.80 ± 2.23	75.73 ± 1.72	64.41
	MedCPT	73.09 ± 1.34	66.69 ± 1.32	54.94 ± 0.77	66.40 ± 1.11	85.76 ± 1.41	69.38
	RRF-2	75.57 ± 1.30	64.34 ± 1.34	55.34 ± 0.77	69.00 ± 1.15	87.06 ± 1.35	70.26
	RRF-4	73.37 ± 1.34	64.73 ± 1.34	54.75 ± 0.77	67.20 ± 1.15	88.51 ± 1.38	69.71
StatPearls (301.2k)	BM25	71.63 ± 1.34	65.67 ± 1.33	54.89 ± 0.77	27.60 ± 2.00	60.36 ± 1.97	56.03
	Contriever	73.28 ± 1.34	67.48 ± 1.33	54.24 ± 0.77	28.80 ± 2.03	58.41 ± 1.95	56.44
	SPECTER	73.74 ± 1.33	64.73 ± 1.34	52.83 ± 0.77	23.20 ± 2.00	57.77 ± 1.95	54.45
	MedCPT	72.82 ± 1.35	64.89 ± 1.34	54.17 ± 0.77	27.60 ± 2.00	60.68 ± 1.96	56.03
	RRF-2	72.64 ± 1.35	65.67 ± 1.33	54.63 ± 0.77	30.00 ± 2.05	61.17 ± 1.96	56.82
	RRF-4	73.83 ± 1.33	65.12 ± 1.34	53.81 ± 0.77	30.60 ± 2.06	59.71 ± 1.97	56.61
Textbooks (125.8k)	BM25	74.66 ± 1.32	66.54 ± 1.32	54.05 ± 0.77	30.20 ± 2.05	60.03 ± 1.97	57.10
	Contriever	74.10 ± 1.33	67.16 ± 1.32	54.53 ± 0.77	26.60 ± 1.98	60.19 ± 1.97	56.52
	SPECTER	72.82 ± 1.35	67.40 ± 1.31	53.29 ± 0.77	25.60 ± 1.95	55.50 ± 1.95	54.92
	MedCPT	74.93 ± 1.31	66.22 ± 1.33	54.41 ± 0.77	29.20 ± 2.03	61.33 ± 1.96	57.22
	RRF-2	76.68 ± 1.28	65.91 ± 1.33	54.79 ± 0.77	31.00 ± 2.07	59.39 ± 1.95	57.55
	RRF-4	75.76 ± 1.30	66.06 ± 1.33	55.56 ± 0.77	30.40 ± 2.06	60.68 ± 1.96	57.69
Wikipedia (29.9M)	BM25	73.37 ± 1.34	63.47 ± 1.35	54.10 ± 0.77	26.40 ± 1.97	71.36 ± 1.82	57.74
	Contriever	74.10 ± 1.33	65.99 ± 1.33	54.03 ± 0.77	26.40 ± 1.97	69.90 ± 1.85	58.08
	SPECTER	72.18 ± 1.36	63.63 ± 1.35	52.71 ± 0.77	22.20 ± 1.95	66.83 ± 1.89	55.51
	MedCPT	71.99 ± 1.36	65.12 ± 1.34	55.15 ± 0.77	29.00 ± 2.03	73.46 ± 1.78	58.95
	RRF-2	74.20 ± 1.33	64.57 ± 1.34	54.72 ± 0.77	31.00 ± 2.07	76.21 ± 1.71	60.14
	RRF-4	73.19 ± 1.34	64.96 ± 1.34	54.53 ± 0.77	31.00 ± 2.07	72.01 ± 1.81	59.14
MedCorp (65.3M)	BM25	73.65 ± 1.34	65.91 ± 1.33	56.78 ± 0.77	66.20 ± 1.15	87.70 ± 1.35	70.05
	Contriever	75.48 ± 1.30	64.10 ± 1.34	56.11 ± 0.77	62.40 ± 2.17	84.95 ± 1.44	68.61
	SPECTER	74.38 ± 1.32	65.44 ± 1.33	54.41 ± 0.77	55.80 ± 2.22	73.14 ± 1.78	64.63
	MedCPT	74.75 ± 1.32	67.40 ± 1.31	55.85 ± 0.77	66.40 ± 1.11	85.92 ± 1.40	70.06
	RRF-2	73.74 ± 1.33	67.24 ± 1.32	56.08 ± 0.77	67.80 ± 1.15	88.19 ± 1.30	70.61
	RRF-4	75.48 ± 1.30	66.61 ± 1.32	58.04 ± 0.77	67.40 ± 1.15	90.29 ± 1.32	71.57

Inference-time knowledge problem

However, BEWARE . . .

The **corpus** as well as the **retrieval approach** can affect performance

Regarding how corpus can affect RAG, look at how performance for PubMedQA improves when the documents used for RAG are from PubMed.

		MIRAGE Benchmark Dataset					Average
Corpus	Retriever	MMLU-Med	MedQA-US	MedMCQA	PubMedQA*	BioASQ-Y/N	
None	None	72.91 ± 1.35	65.04 ± 1.34	55.25 ± 0.77	36.00 ± 2.15	74.27 ± 1.76	60.69
PubMed (23.9M)	BM25	72.27 ± 1.36	63.71 ± 1.35	55.49 ± 0.77	66.20 ± 1.15	88.51 ± 1.38	69.23
	Contriever	71.72 ± 1.35	63.94 ± 1.35	54.29 ± 0.77	65.60 ± 1.15	85.44 ± 1.42	68.20
	SPECTER	73.19 ± 1.34	65.20 ± 1.34	53.12 ± 0.77	54.80 ± 2.23	75.73 ± 1.72	64.41
	MedCPT	73.09 ± 1.34	66.69 ± 1.32	54.94 ± 0.77	66.40 ± 1.15	85.76 ± 1.41	69.38
	RRF-2	75.57 ± 1.30	64.34 ± 1.34	55.34 ± 0.77	69.00 ± 1.15	87.06 ± 1.35	70.26
	RRF-4	73.37 ± 1.34	64.73 ± 1.34	54.75 ± 0.77	67.20 ± 1.15	88.51 ± 1.38	69.71
StatPearls (301.2k)	BM25	71.63 ± 1.35	65.67 ± 1.33	54.89 ± 0.77	27.60 ± 2.00	60.36 ± 1.97	56.03
	Contriever	73.28 ± 1.34	67.48 ± 1.33	54.24 ± 0.77	28.80 ± 2.03	58.41 ± 1.95	56.44
	SPECTER	73.74 ± 1.33	64.73 ± 1.34	52.83 ± 0.77	23.20 ± 2.02	57.77 ± 1.92	54.45
	MedCPT	72.82 ± 1.35	64.89 ± 1.34	54.17 ± 0.77	27.60 ± 2.00	60.68 ± 1.96	56.03
	RRF-2	72.64 ± 1.35	65.67 ± 1.33	54.63 ± 0.77	30.00 ± 2.05	61.17 ± 1.96	56.82
	RRF-4	73.83 ± 1.33	65.12 ± 1.34	53.81 ± 0.77	30.60 ± 2.06	59.71 ± 1.97	56.61
Textbooks (125.8k)	BM25	74.66 ± 1.32	66.54 ± 1.32	54.05 ± 0.77	30.20 ± 2.05	60.03 ± 1.97	57.10
	Contriever	74.10 ± 1.33	67.16 ± 1.32	54.53 ± 0.77	26.60 ± 1.98	60.19 ± 1.97	56.52
	SPECTER	72.82 ± 1.35	67.40 ± 1.31	53.29 ± 0.77	25.60 ± 1.95	55.50 ± 1.93	54.92
	MedCPT	74.93 ± 1.31	66.22 ± 1.33	54.41 ± 0.77	29.20 ± 2.03	61.33 ± 1.96	57.22
	RRF-2	76.68 ± 1.29	65.91 ± 1.33	54.79 ± 0.77	31.00 ± 2.07	59.39 ± 1.95	57.55
	RRF-4	75.76 ± 1.30	66.06 ± 1.33	55.56 ± 0.77	30.40 ± 2.06	60.68 ± 1.96	57.69
Wikipedia (29.9M)	BM25	73.37 ± 1.34	63.47 ± 1.35	54.10 ± 0.77	26.40 ± 1.97	71.36 ± 1.82	57.74
	Contriever	74.10 ± 1.33	65.99 ± 1.33	54.03 ± 0.77	26.40 ± 1.97	69.90 ± 1.85	58.08
	SPECTER	72.18 ± 1.36	63.63 ± 1.35	52.71 ± 0.77	22.20 ± 1.95	66.83 ± 1.89	55.51
	MedCPT	71.99 ± 1.36	65.12 ± 1.34	55.15 ± 0.77	29.00 ± 2.03	73.46 ± 1.78	58.95
	RRF-2	74.20 ± 1.33	64.57 ± 1.34	54.72 ± 0.77	31.00 ± 2.07	76.21 ± 1.71	60.14
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	Contriever	75.48 ± 1.30	64.10 ± 1.34	56.11 ± 0.77	62.40 ± 2.17	84.95 ± 1.44	68.61
	SPECTER	74.38 ± 1.32	65.44 ± 1.33	54.41 ± 0.77	55.80 ± 2.22	73.14 ± 1.78	64.63
	MedCPT	74.75 ± 1.32	67.40 ± 1.31	55.85 ± 0.77	66.40 ± 1.15	85.92 ± 1.40	70.06
	RRF-2	73.74 ± 1.33	67.24 ± 1.32	56.08 ± 0.77	67.80 ± 1.15	88.19 ± 1.30	70.61
	RRF-4	75.48 ± 1.30	66.61 ± 1.32	58.04 ± 0.77	67.40 ± 1.15	90.29 ± 1.32	71.57

Inference-time knowledge problem

However, BEWARE . . .

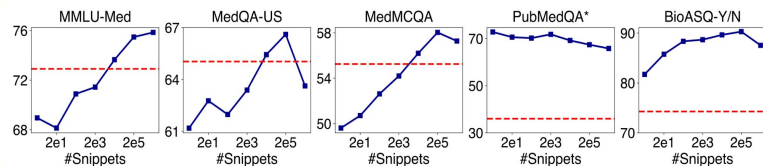
The **corpus** as well as the **retrieval approach** can affect performance

Regarding how corpus can affect RAG, look at how performance for PubMedQA improves when the documents used for RAG are from PubMed.

Also look at how combining all of the documents into one corpora led to an overall jump in performance across the benchmark

Corpus	Retriever	MIRAGE Benchmark Dataset					Average
		MMLU-Med	MedQA-US	MedMCQA	PubMedQA*	BioASQ-Y/N	
None	None	72.91 ± 1.35	65.04 ± 1.34	55.25 ± 0.77	36.00 ± 2.15	74.27 ± 1.76	60.69
PubMed (23.9M)	BM25	72.27 ± 1.36	63.71 ± 1.35	55.49 ± 0.77	66.20 ± 1.15	88.51 ± 1.38	69.23
	Contriever	71.72 ± 1.35	63.94 ± 1.35	54.29 ± 0.77	65.60 ± 1.15	85.44 ± 1.42	68.20
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Inference-time knowledge problem



Also beware...

Retrieving more chunks can help up to a point, but too many can introduce noise and hurt performance.

Inference-time knowledge problem

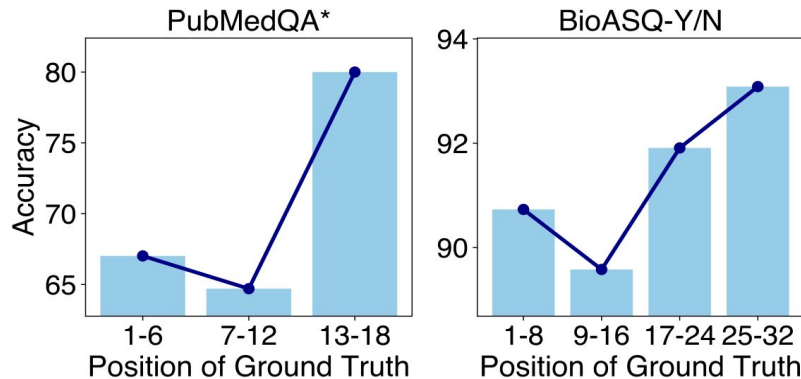
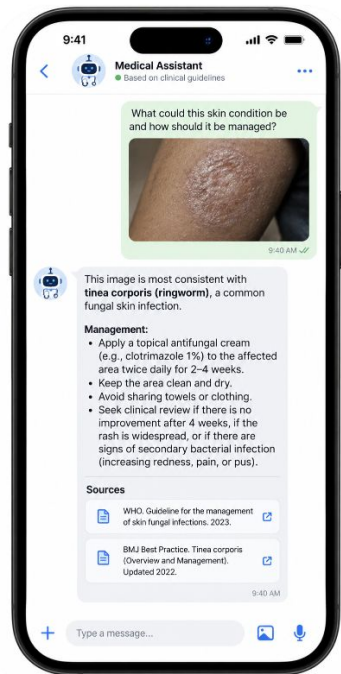


Figure 4: The relations between QA accuracy and the position of the ground-truth snippet in the LLM context.

And also ...

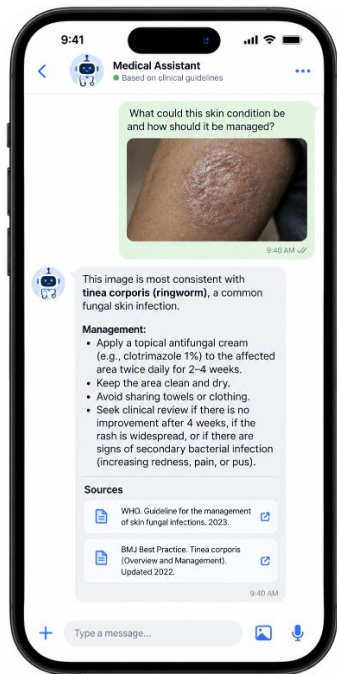
The position of useful chunks in top-k chunks returned matter

Output verification problem



Even after training the model better and giving it access to more information, can we verify what it actually says?

Output verification problem



Even after training the model better and giving it access to more information, can we verify what it actually says?

MedScore: Generalizable Factuality Evaluation of Open-ended Long-form Medical Answers by Domain-adapted Claim Decomposition and Verification

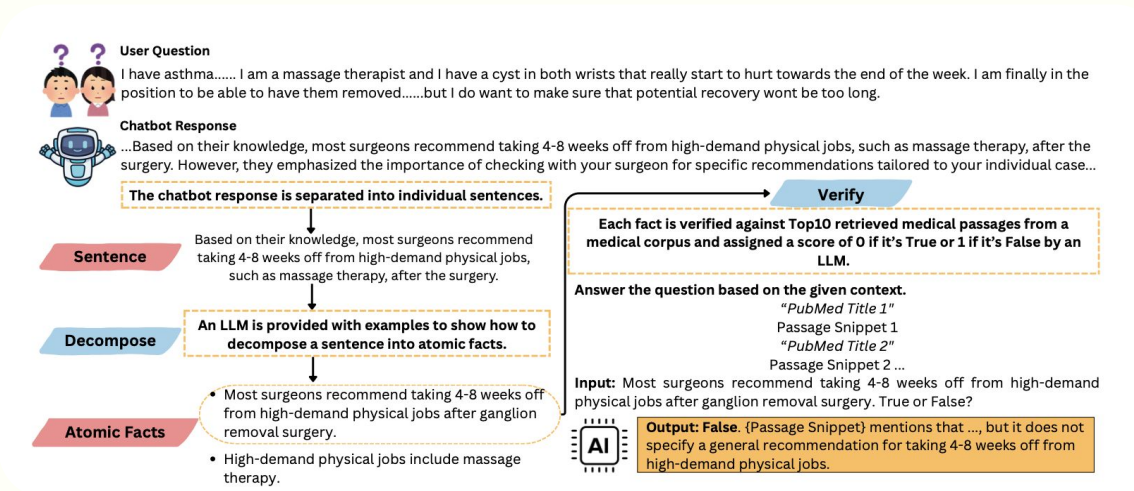
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Baltimore, MD 21218, USA
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Abstract

While Large Language Models (LLMs) can generate fluent and convincing responses, they are not necessarily correct. This is especially apparent in the popular decompose-then-verify factuality evaluation pipeline, where LLMs evaluate generated text by decomposing it into individual, valid claims. Factuality evaluation is especially important for medical answers, since incorrect medical information could seriously harm the patient. However, existing factuality systems are a poor match for the medical domain, as they are typically only eval-

into “atomic facts” and then *verified* against a knowledge source (Min et al., 2023). The final “factuality” score is the proportion of claims that can be verified. The popular FActScore framework (Min et al., 2023) exemplifies this approach, which uses LLMs to both decompose and verify claims. Subsequent work has shown the importance of the decomposition step, as the ability to correctly decompose a generation into atomic facts directly influences the resulting factuality score (Wanner et al., 2024). Existing frameworks have been evaluated almost exclusively on biographical data, where

Output verification problem



Questions?

Thank You

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